

COURTSIDE: Pricing the Value of Speed in In-Play Tennis Prediction Markets

[Author names: team to fill] · University of Florida · [emails: team to fill]
Gator Quant Hacks 2026 · Systematic Trading Track · University of Florida
October 4, 2026

Abstract. We study who profits from speed in Polymarket’s in-play tennis moneylines, using public trade tapes for 13,084 matches. A walk-forward fast tier of wallets trading within 3 s of a score move earns +0.93¢ per share after fees in sample and +0.75¢ out of sample (9/9 and 3/3 months positive); other takers lose, and copying its trades 3 s later loses. At the fast tier’s own fills a frozen book earns +0.60¢ per share out of sample (Sharpe 6.7) but turns negative when fees double. Simulated at a 1 s licensed-feed baseline (assumed feed latency (licensed feed not purchased); parameters measured), a computer-vision (CV) trader earns +\$57 a day out of sample under a calibrated, post hoc stamp lag and +\$4 under the pre-registered one, which breaks even at 1.0–1.1 s of feed delay. Real money: none.

JEL: G14, G12, C58. **Keywords:** prediction markets; latency arbitrage; in-play betting; tennis; computer vision; market microstructure.

1 Summary

In-play tennis prediction markets are priced by whoever learns the point first. We measure who that is, what they earn and what each second of speed is worth; our CV calls the point before the ball lands, and we price the feed it needs.

-
- **Strategy.** Call the point with CV before the ball lands; take the stale quote inside the 1 s taker delay. CV results: assumed feed latency (licensed feed not purchased); parameters measured.
 - **Edge.** The book reprices 1.2 s *before* the official stamp. The fast tier earns +0.93 / +0.75¢ per share after fees (IS / OOS) every month; other takers lose −0.89 / −1.43¢.
 - **Headline OOS.** At fast-tier fills: +0.60¢ [0.09, 1.13], Sharpe 6.7. At the 1 s baseline: +\$57/day (Sharpe 8.8; calibrated, post hoc) or +\$4/day (Sharpe 0.3; pre-registered); break-even delay 2.1–2.2 or 1.0–1.1 s, so public streams and score feeds lose.
 - **What failed.** Fees ×2 OOS (−0.34¢); every blind test of a tradable book; the live-book replay; table tennis; OOS after a central feed licence (−\$75/day). Real money: none.
-

2 Economic hypothesis

Win probability is a known function of the score. The leverage of point k ,

$$\Delta(k) = \Pr(\text{win} \mid \text{point } k \text{ won}) - \Pr(\text{win} \mid \text{point } k \text{ lost}), \quad (1)$$

follows from the exact point-level Markov chain (Klaassen and Magnus, 2001), with mean $|\Delta(k)|$ of 5.7% over the 161 points of a simulated ATP best-of-three. Each point moves fair value by a known

Reproduce: bash reproduce.sh (<https://github.com/OjasMishra32/computervisionpredictivetabletennismoneymachineouuushiii>). Real money: none (Participant Terms 5.3). CV-strategy results: assumed feed latency (licensed feed not purchased); parameters measured. We hold no licensed feed, no match footage and no camera at any court.

amount at a known instant, and the news reaches traders in tiers (Fig. 1a): in-venue tracking; the official stamp, an unmeasured 1–3 s later; licensed video, 0.5–8 s (vendor-stated); public streams, ≥ 12.2 s (preliminary); score feeds, 28.2–44.1 s behind the book, which reprises a median 1.2 s *before* the stamp ($n = 482$).

H6. *Universe:* Polymarket ATP/WT/Challenger moneylines. *Behaviour:* wallets trading 0–3 s after a score move. *Horizon:* 30 s markout. *Counterparty:* slower takers on an older tier. *Persistence:* physics, data licensing, the 1 s taker delay. *Prediction:* fast wallets chosen on past months keep a positive net markout each month and out of sample. *Falsifier:* ≤ 0 in over a third of months, or out of sample. *Trail:* H1–H4 committed before any result (7232986, 2026-10-03 09:50 UTC); H6 written in sample and frozen before the OOS opened; v2 in HYPOTHESIS_V2.md; changes in DEVIATIONS.md.

The edge is a structural constraint, not a risk premium, and tracks the structure: v2 earns +1.96, +1.36, +1.45 and +1.02¢ per share in the 3 s/0%, 3 s/3%, 1 s/3% and 1 s/5% delay/fee regimes. The fast tier grew from 4 to 131 wallets; its edge shrank 0.20¢ a month ($t = -3.98$) but stayed positive. Latency races (Budish et al., 2015; Aquilina et al., 2022) and in-play mispricing (Croxxson and Reade, 2014; Brown, 2012, 2014) are known; new here: a wallet-level map of the latency tier, a timing of every public source, a price on each second.

3 Data and universe

13,084 resolved ATP/WT/Challenger singles moneylines with $\geq \$5,000$ volume, Oct 8, 2025 to Oct 3, 2026: binary contracts on a public order book with free trade data. IS is the first 80% of matches by start time; OOS the last 2,617 from 2026-08-25 14:15 UTC (21.2% of volume; burned for v2). A blind test on 11,307 smaller markets checks the volume filter’s survivorship. 6 tapes may be truncated (kept); 2,889 early matches without a fee schedule pay 0%; 2.89% settle 50/50 (kept); tape stamps lag 1.98 s. Sources: Polymarket APIs and sports websocket (Polymarket, 2026); ESPN and WTA, latency only (ESPN, 2026; WTA, 2026); Kalshi (Kalshi, 2026); French (2026); OpenTTGames (Voeikov et al., 2020); BlurBall and WASB-SBDT weights (Gossard et al., 2026; Tarashima et al., 2023).

4 Methodology

Fast tier and v2. A detector fires on a ≥ 4 ¢ move of the 10 s VWAP against the prior 60 s. Each month a wallet qualifies on past months only (≥ 30 prints 0–3 s after a detection, ≥ 10 matches, $t > 3$ on its 30 s markout). The frozen book v2 copies qualifying prints, sized $\propto 1/\sqrt{q(1-q)}$, at most \$1,000 per order and 100 shares net per match, held to resolution, at the fast tier’s own prices: *the opportunity at their speed, not our execution*.

CV strategy. Our tracker gets 11 of 11 out-calls right 50 ms before contact on held-out 120 fps table-tennis video (Fig. 1b) and streams 102,120 frames at 119.9 fps on one GPU with 0 drops. With the bounce at time zero, an order wins only if

$$V + \delta + D - \ell < L + R \iff V < L + R - D - \delta + \ell, \quad (2)$$

where V is the feed delay, δ the inference (20 ms), network and gateway delays, D the 1 s taker delay, ℓ the CV call lead, L the stamp lag and R the reprice relative to the stamp. R , depth, network and fees are measured on 482 live points; V and L are assumed: V runs 0–60 s (20 seeds per cell) and the unmeasured L gets two readings, pre-registered 2.0 s and calibrated, post hoc 3.14 s.

Costs. Each match’s own taker fee,

$$f(q) = r q (1 - q) \text{ per share} = 10,000 r (1 - q) \text{ bps of notional}, \quad (3)$$

with $r = 5\%$ today (250 bps at $q = 0.5$): 120/179 bps on v2’s trades (IS/OOS). Spreads are paid as traded. Stresses: $\frac{1}{2}$ and 1 tick worse entry, fees $\times 2$, all costs $\times 2$ ($\approx +99$ bps more). **Validation.** Monthly walk-forward inside IS; past-month qualification; the fee in force; the onset look-ahead fixed (D9); every OOS read logged.

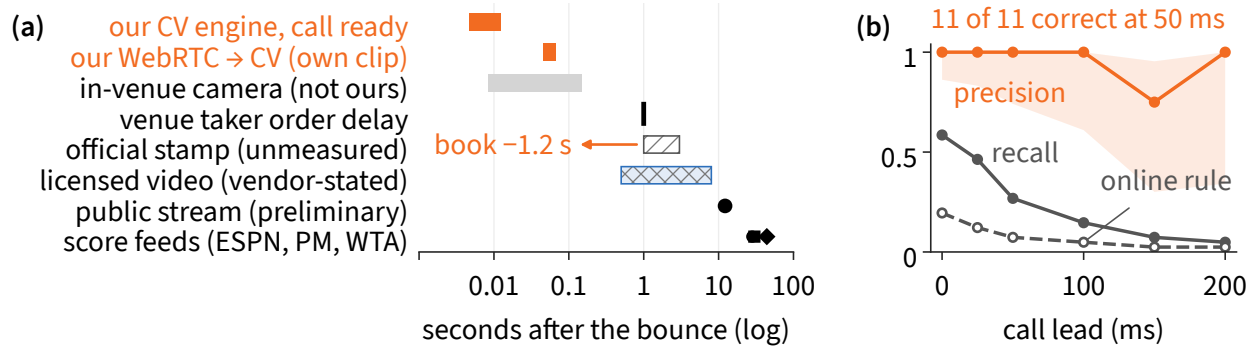


Figure 1. The race is decided inside the venue's 1 s taker delay; only in-venue information beats the book. (a) Delay after the bounce (log): orange = ours (WebRTC: our own clip, loopback, 46 ms p50, 67 ms p99); hatched = vendor-stated or model range. (b) Early out-calls, held-out 120 fps table-tennis video (41 out, 130 in; band: Wilson 95%). Source: Polymarket and WTA feeds; results/{tracking,engine,webrtc}.

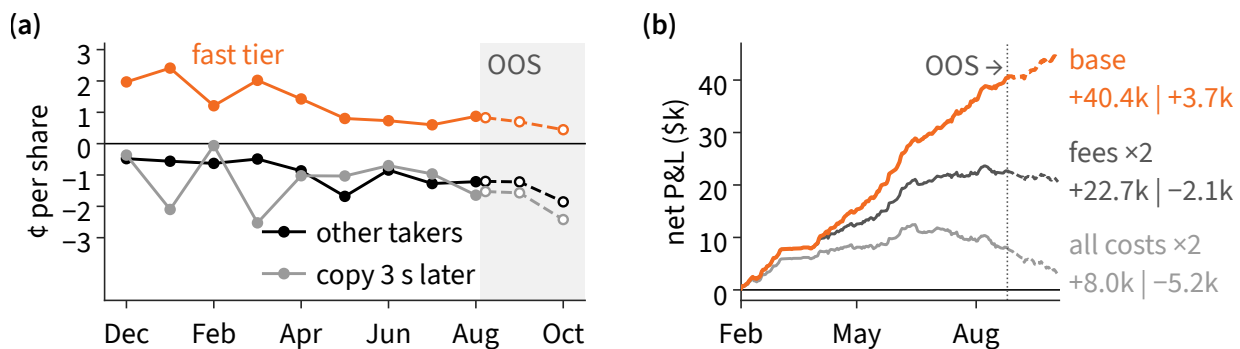


Figure 2. Speed is the edge. (a) Monthly net 30 s markout after fees; filled = IS, hollow = OOS. (b) v2 net P&L at the fast tier's own fills (the opportunity at their speed, not our execution), base and costs doubled; solid = IS, dashed = OOS; labels: IS | OOS totals. Source: results/{alpha,lowloss,v2}.

5 Results

The fast tier beats other takers every month and copying it 3 s later loses (Fig. 2a). At its fills v2 earns in both periods (Table 1, Fig. 2b); out of sample the edge is gone at $\frac{1}{2}$ tick worse entry (+0.10¢) and negative with fees doubled (−0.34¢, 0/3 months positive). **Sharpe above 3.** We assumed a bug and found one: fixing the onset look-ahead (D9) cut IS Sharpe from 16.8 to 14.5. The rest is about 270 small, nearly independent bets a day (same-day correlation 0.0013). The deflated Sharpe (Bailey and López de Prado, 2014) is 0.997 IS at $N = 3,386$ but 0.075–0.839 OOS: 40 days cannot rule out luck. The CV Sharpe is a simulation's seed mean and ignores model risk. **The 1 s baseline.** Table 2 and Fig. 3 give the main result. The two readings bracket it: +\$57/day OOS (Sharpe 8.8) calibrated against +\$4/day (Sharpe 0.3, −0.38¢ per share) pre-registered. Public streams and score feeds sit far right of break-even; under the stamp-noise reading the trader loses at every V (Table A1). A replay of 9 matches recorded live on 2026-10-03 against their real books loses in 36 of 36 settings (−0.94¢ at $V = 1$ s; Fig. 3d): it trades every point, not only ≥ 4 ¢ jumps; one day, an illustration. Every blind test of a tradable book failed (Table 3); the fast-minus-others gap stayed positive wherever measured.

6 Risk management

- **Limits.** \$1,000 per order, 100 shares net per match, zone 0.05–0.95; capital 3× peak locked (\$28,302); worst day −\$551 / −\$469 (IS / OOS). Kill switches in code (engine/risk/): daily stop \$1,000 (3.86 σ), stale feed >2 s or vision >1 s, latency above its rolling p95.

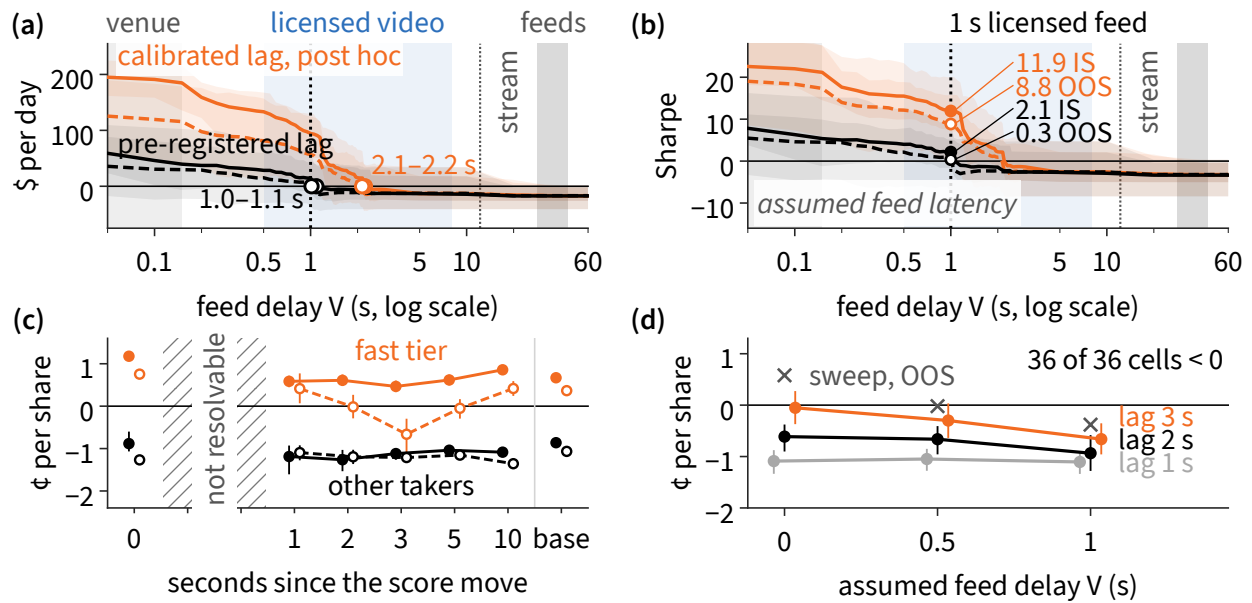


Figure 3. Signal decay: each second of feed latency costs the edge. (a) \$/day, (b) Sharpe against feed delay V for both stamp lags; solid = IS, dashed = OOS; bands: 20 seeds; circles: break-even. (c) Net markout by seconds since the move. (d) Replay of 9 live-recorded matches on real books; $\times$: sweep. Simulated at an assumed feed latency (licensed feed not purchased); parameters measured. Source: results/{tier0,decay,replay}.

- **De-risking, set in advance.** Halve size if the trailing 30-day edge is below 0.3¢ (IS minimum +0.74¢); stop at ≤ 0 , at a 5% drawdown, or on a fee or delay change.
- **Largest risks.** Venue rules (doubled fees erase the OOS edge); crowding (top five wallets: 81%/142% of P&L); tails (kurtosis 4.30/3.26); legal (courtsiding breaches ticket terms, [International Tennis Federation, 2026](#)). Factors (Fama and French, 1993; Carhart, 1997): $\alpha t = 8.48$, $\max |t| = 1.36$, R^2 3.3% (IS; Table A3).

Table 1. v2 earns in both periods at the fast tier's own fills; its OOS edge does not survive doubled fees.

	In sample Feb 1–Aug 25, 206 d	Burned OOS Aug 25–Oct 3, 40 d
Trades (matches)	55,662 (7,639)	10,412 (2,003)
Net ¢ per share [95% CI]	+1.38 [1.17, 1.59]	+0.60 [0.09, 1.13]
Net edge / fee, bps of notional	273 / 120	114 / 179
Annualised return / volatility	253% / 17.5%	148% / 22.2%
Sharpe, annualised [bootstrap 95% CI]	14.5 [11.9, 17.4]	6.7 [1.9, 12.3]
Max drawdown / worst day, % of capital	–2.0% / –1.9%	–2.1% / –2.1%
Skew / worst month	+0.53 / +\$2,543	+0.33 / –\$434
Turnover ($\times$ capital a year) / months positive	93 / 7/7	129 / 2/3
Costs stressed: $\frac{1}{2}$ / 1 tick worse entry	+0.88 / +0.38	+0.10 / –0.40
Fees $\times 2$ [CI] (months positive)	+0.77 [0.57, 0.98] (7/7)	–0.34 [–0.86, 0.19] (0/3)
All costs $\times 2$ [CI]	+0.27 [0.07, 0.48]	–0.84 [–1.36, –0.31]

Notes: held to resolution, net of fees and spreads; capital $3\times$ peak locked (\$28,302 / \$22,754); CIs by match (by wallet, OOS: [–0.45, 2.38]). Source: results/v2/.

Table 2. Main result: simulated at a 1 s licensed-feed baseline; assumed feed latency (licensed feed not purchased); parameters measured.

At $V = 1$ s	Calibrated $L = 3.14$ s (post hoc)		Pre-registered $L = 2.0$ s	
	IS	OOS	IS	OOS
Net \$ per day [seed 95%]	+\$94 [51, 125]	+\$57 [1, 130]	+\$15 [-9, 43]	+\$4 [-36, 62]
Net ¢ per share	+1.11	+0.66	+0.40	-0.38
95% CI	[0.85, 1.38]	[0.10, 1.19]	[-0.32, 1.08]	[-2.08, 1.21]
Sharpe, annualised	11.9	8.8	2.1	0.3
Ann. return / vol, %	118 / 9.9	91 / 10.3	30 / 14.1	11 / 33.4
Max DD % / worst day	-1.6 / -\$223	-1.6 / -\$192	-6.6 / -\$263	-5.3 / -\$228
Break-even V , s [95% CI]	2.23	2.14	1.09	1.01
	[2.18, 3.39]	[1.59, 2.19]	[1.03, 2.04]	[0.43, 1.05]

Notes: means of 20 draws; IS: 1 s-delay matches (103 d); OOS burned (40 d); vol = return/Sharpe; 24–65 trades a day; no daily path stored (equity curve: v2). At $L = 1$ s: -\$6 / -\$11 a day.

7 Liquidity and capacity

v2 trades \$7,182 a day, 0.07% of match volume. Its OOS per-share CI excludes zero only up to 1× size (\$22,754 of capital); 2× spans zero and 5× loses (Fig. A1a). Median stale depth before a reprice: \$222, \$379, \$564 at 0, 1, 2 s; spread 1¢. Fees bind: the OOS break-even fee is 8.2%, or 2.4% after an assumed \$5,000-a-month feed licence, which takes the 1 s-baseline OOS P&L to -\$110 (calibrated) or -\$163 (pre-registered) a day (Fig. A1b).

8 Limitations and next steps

The key uncertainty is the stamp lag L (Eq. 2), which moves the break-even one for one; one live session with a licensed low-latency feed, timing bounce against stamp, would measure it. Licensing that feed turns the simulation into deployment (Polymarket US already licenses official data and live streams: Genius Sports for selected leagues, ATP Tour streaming rights; [Gaming Intelligence, 2026](#); [PR Newswire, 2026](#)); a faster feed moves left on the curve (Fig. 3a); then an organiser-consented venue camera. Risks: fee rises, a longer delay, faster rivals, wallet exits, legal access. Table tennis is illiquid (median spread 94¢).

Table 3. Every blind test of a tradable book failed or is pending.

Test, ¢ per share	IS	OOS or blind	Verdict
H1 · H2 · H5 (post hoc)	-1.61 · -0.41 · +0.22	-2.08 · +0.80 · +0.16	fail · none · n.s.
v1 copy fast tier (OOS opened once)	+1.16	+0.57 (-\$36,056)	fail
v2, 11,307 unseen markets (blind)	+2.02	+1.22 [-0.19, 2.65]	fail
Tier-0 v3 rule, unseen markets (blind)	-0.22	-0.52	fail
Maker v1 (blind), per fill	+2.02	+1.87 [-0.14, 3.86]	fail, -\$379
v2 after fixed costs, \$ a day	+\$29	-\$75	fail OOS
Table tennis TT1–TT5, 27 matches	no fast wallet qualifies		untestable
Replay on live books, $V = 1$ s	-0.94 [-1.28, -0.58]		loses
Forward test · live paper session			pending · pending

Notes: 4,219 variants tried (Table A4). Fast minus others, unseen markets: +3.09 / +2.16¢ (the latency gap held). Held-out data read 69 times (7 blind first runs, 13 non-blind, 5 descriptive, 22 audits, 20 replay/live, 2 other; Table A7); rules changed after a look 2 times.

References

- Ably (n.d.). Case study: Genius Sports. <https://ably.com/case-studies/genius-sports>.
- Aquilina, M., Budish, E., and O'Neill, P. (2022). Quantifying the high-frequency trading “arms race”. *Quarterly Journal of Economics*, 137(1):493–564.
- Bailey, D. H., Borwein, J., López de Prado, M., and Zhu, Q. J. (2017). The probability of backtest overfitting. *Journal of Computational Finance*, 20(4):39–69.
- Bailey, D. H. and López de Prado, M. (2014). The deflated Sharpe ratio: Correcting for selection bias, backtest overfitting, and non-normality. *Journal of Portfolio Management*, 40(5):94–107.
- Brown, A. (2012). Evidence of in-play insider trading on a UK betting exchange. *Applied Economics*, 44(9). Cited from its abstract.
- Brown, A. (2014). Information processing constraints and asset mispricing. *Economic Journal*, 124(575).
- Budish, E., Cramton, P., and Shim, J. (2015). The high-frequency trading arms race: Frequent batch auctions as a market design response. *Quarterly Journal of Economics*, 130(4):1547–1621.
- Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52(1):57–82.
- Croxson, K. and Reade, J. J. (2014). Information and efficiency: Goal arrival in soccer betting. *Economic Journal*, 124(575):62–91.
- ESPN (2026). Public tennis scoreboard endpoint. Read live on 2026-10-03 for latency timing only; derived timings committed.
- Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.
- French, K. R. (2026). Data library: Fama–French factors and momentum. https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html.
- Gaming Intelligence (2026). Genius sports agrees Polymarket prediction market deal. <https://www.gamingintelligence.com/finance/234424-genius-sports-agrees-polymarket-prediction-market-deal/>. 5 August 2026; Polymarket US, selected leagues.
- GL Systemhaus (2018). Who cheers first? About latencies in sports livestreaming. <https://www.gl-systemhaus.de/en/blog/who-cheers-first-about-latencies-in-sports-livestreaming>.
- Gossard, T., Radovic, F., Ziegler, A., and Zell, A. (2026). BlurBall: Joint ball and motion blur estimation for table tennis ball tracking. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*. arXiv:2509.18387; code and weights MIT.
- Harvey, C. R., Liu, Y., and Zhu, H. (2016). ...and the cross-section of expected returns. *Review of Financial Studies*, 29(1):5–68.
- Harvey, C. R., Rattray, S., Sinclair, A., and Van Hemert, O. (2017). Man vs. machine: Comparing discretionary and systematic hedge fund performance. *Journal of Portfolio Management*, 43(4):55–69.
- Huang, Y.-C., Liao, I.-N., Chen, C.-H., İk, T.-U., and Peng, W.-C. (2019). TrackNet: A deep learning network for tracking high-speed and tiny objects in sports applications. In *IEEE International Conference on Advanced Video and Signal Based Surveillance (AVSS)*.
- International Tennis Federation (2026). ITF world tennis tour regulations 2026.
- International Tennis Integrity Agency (2026). Tennis anti-corruption program 2026.
- ISPreview (2021). Broadcast lag in live online TV sport streaming frustrates fans. <https://www.ispreview.co.uk/index.php/2021/06/broadcast-lag-in-live-online-tv-sport-streaming-frustrates-fans.html>.
- Kalshi (2026). Public market-data API. <https://trading-api.readme.io>. Accessed October 2026.

- Klaassen, F. J. G. M. and Magnus, J. R. (2001). Are points in tennis independent and identically distributed? Evidence from a dynamic binary panel data model. *Journal of the American Statistical Association*, 96(454):500–509.
- NEXT.io (n.d.). Matt fleckenstein: raising the bar in in-play betting (Genius BetVision). <https://next.io/news/betting/matt-fleckenstein-raising-bar-in-play-betting/>. Vendor statement, 4–8 s venue to phone; unverified.
- Politis, D. N. and Romano, J. P. (1994). The stationary bootstrap. *Journal of the American Statistical Association*, 89(428):1303–1313.
- Polymarket (2026). Gamma, data and CLOB APIs, sports websocket, fee and rate-limit documentation. <https://docs.polymarket.com>. Public, no keys; accessed October 2026.
- PR Newswire (2026). Polymarket secures exclusive ATP Tour streaming rights for prediction markets. <https://www.prnewswire.com/news-releases/polymarket-secures-exclusive-atp-tour-streaming-rights-for-prediction-markets-302841534.html>. 3 August 2026.
- Regen Sports (2026). I attended Sportradar’s game, set, tech. <https://regensports.substack.com/p/i-attended-sportradars-game-set-tech>.
- Sportradar (n.d.). Live streams for betting operators. <https://sportradar.com/betting-gaming/products/live-streams/>. Vendor page.
- Stats Perform (n.d.). Realtime streaming; official WTA data for sportsbooks. <https://www.statsperform.com/products/realtime-streaming/>. Vendor claim, unverified.
- Tarashima, S. et al. (2023). Widely applicable strong baseline for sports ball detection and tracking. In *British Machine Vision Conference (BMVC)*. WASB-SBDT, MIT.
- The Desk (2026). Stats perform and phenix measure latency at Super Bowl LX. <https://thedesk.net/2026/02/stats-perform-phenix-latency-super-bowl-lx/>.
- Voeikov, R., Falaleev, N., and Baikulov, R. (2020). TNet: Real-time temporal and spatial video analysis of table tennis. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*. OpenTTGames dataset, CC BY-NC-SA 4.0.
- WTA (2026). Public scores and point-by-point API. Read live on 2026-10-03 for latency timing only; derived timings committed.
- Disclosure.** No real money was used. Simulated components are labelled. The authors hold no licensed data feed, no match footage and no court access. University of Florida students; [team to confirm any other disclosures].

Appendix

Everything a judge needs for scoring is in the five main pages; this appendix gives the full grids, logs and tallies behind them. Every appendix float repeats its label.

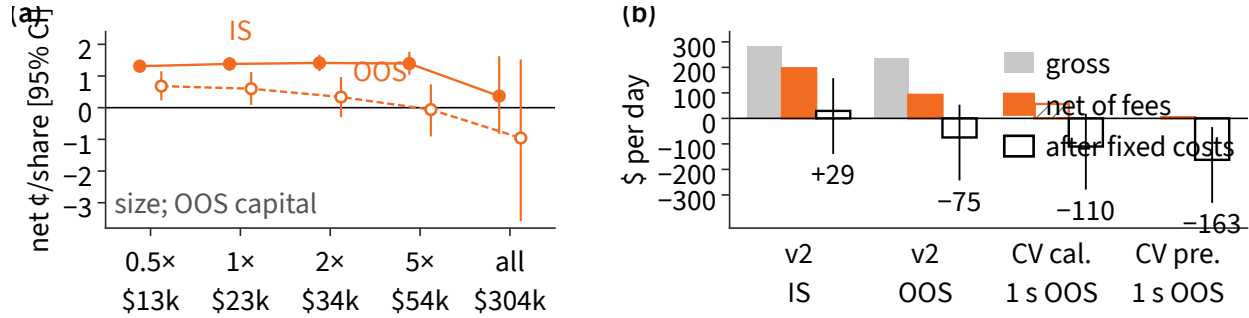


Figure A1. *Capacity is small and costs bind.* (a) v2 net edge by size (caps scaled; optimistic at large size); filled = IS, hollow = OOS. (b) \$/day: gross, net of fees, after fixed costs (low-high); CV: 1 s baseline, OOS, assumed feed latency (licensed feed not purchased); parameters measured. Source: results/financials.

Table A1. *Latency sweep, every reading: Net \$ per day against feed delay V (s), IS and burned OOS.* Simulated: assumed feed latency (licensed feed not purchased); parameters measured.

Stamp lag reading		0	0.5	1	2	5	10	60
calibrated 3.14 s (post hoc)	IS	199	133	94	15	-13	-13	-17
	OOS	129	81	57	7	-13	-13	-17
pre-registered 2.0 s	IS	89	28	15	-6	-13	-14	-17
	OOS	46	15	4	-11	-13	-13	-17
stress 1.0 s	IS	15	-5	-6	-13	-13	-14	-17
	OOS	4	-11	-11	-13	-13	-13	-17
lag 3.0 s	IS	189	121	89	15	-13	-13	-17
	OOS	117	76	46	4	-13	-13	-17
stamp-noise reading	IS	-30	-17	-17	-17	-17	-17	-17
	OOS	-26	-17	-17	-17	-17	-17	-17
stamp-noise, calibrated	IS	199	-45	-17	-17	-17	-17	-17
	OOS	133	-40	-17	-17	-17	-17	-17

Notes: seed means over 20 draws per cell (18,880 runs in all). The stamp-noise readings spread the reprice time by umpire-stamp noise instead of by tournament. Source: results/tier0/latency_sweep.json::video_own120.

Table A2. Latency sweep, every reading: Annualised Sharpe against feed delay V (s), IS and burned OOS. Simulated: assumed feed latency (licensed feed not purchased); parameters measured.

Stamp lag reading		0	0.5	1	2	5	10	60
calibrated 3.14 s (post hoc)	IS	22.9	15.5	11.9	2.2	-2.7	-2.7	-3.4
	OOS	19.6	12.2	8.8	0.9	-2.6	-2.6	-3.2
pre-registered 2.0 s	IS	11.4	4.0	2.1	-1.4	-2.7	-2.9	-3.4
	OOS	7.3	2.1	0.3	-2.3	-2.6	-2.6	-3.2
stress 1.0 s	IS	2.1	-1.3	-1.4	-2.6	-2.7	-2.9	-3.4
	OOS	0.3	-2.3	-2.3	-2.6	-2.6	-2.6	-3.2
lag 3.0 s	IS	21.8	14.4	11.4	2.1	-2.7	-2.7	-3.4
	OOS	17.5	11.4	7.3	0.3	-2.6	-2.6	-3.2
stamp-noise reading	IS	-5.9	-3.4	-3.4	-3.4	-3.4	-3.4	-3.4
	OOS	-5.0	-3.2	-3.2	-3.2	-3.2	-3.2	-3.2
stamp-noise, calibrated	IS	25.1	-8.7	-3.4	-3.4	-3.4	-3.4	-3.4
	OOS	21.2	-7.6	-3.2	-3.2	-3.2	-3.2	-3.2

Notes: seed means over 20 draws per cell (18,880 runs in all). The stamp-noise readings spread the reprice time by umpire-stamp noise instead of by tournament. Source: results/tier0/latency_sweep.json::video_own120.

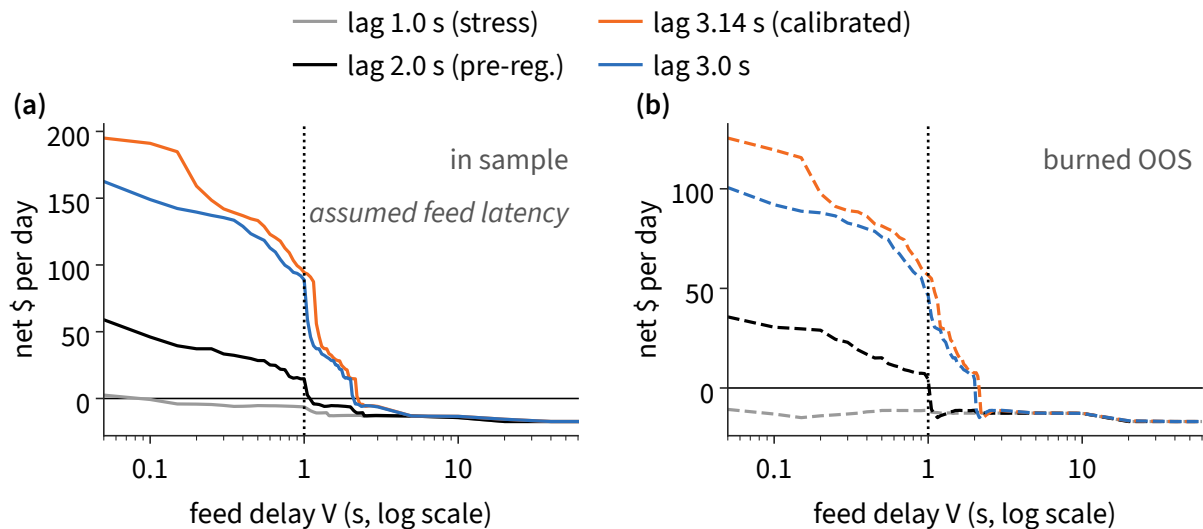


Figure A2. The break-even shifts one for one with the stamp lag. Net \$/day against feed delay for every stamp-lag reading, (a) IS and (b) burned OOS. Simulated: assumed feed latency (licensed feed not purchased); parameters measured. Source: results/tier0/latency_sweep.csv.

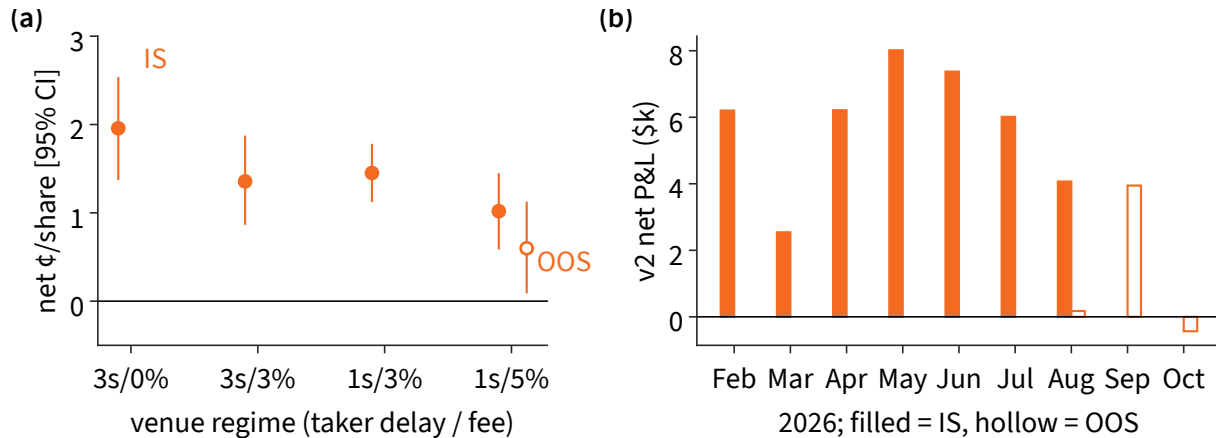


Figure A3. *v2's edge falls as the venue's protection falls, and stays positive in every in-sample regime.* (a) Net ¢ per share by venue regime (taker delay / fee) with 95% CIs; (b) net P&L by month. Measured at the fast tier's own fills: the opportunity at their speed, not our execution. Source: results/risk/risk_stats.json.

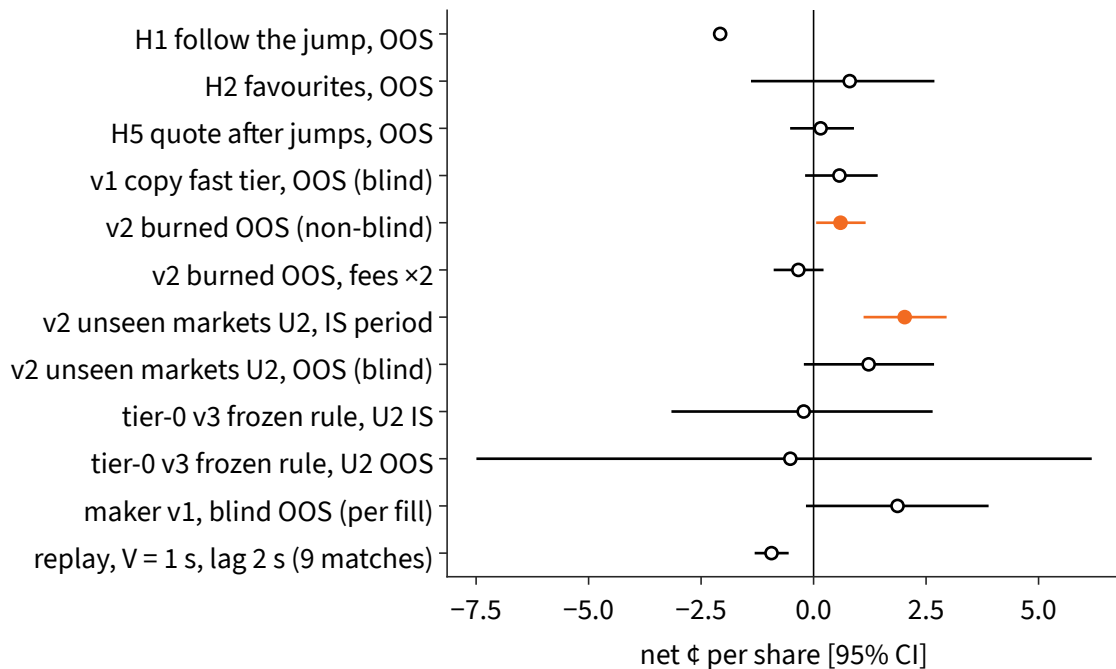


Figure A4. *Every test on one axis.* Point estimates and 95% CIs, net ¢ per share; orange = CI above zero, hollow = the test failed or has no edge. Source: as Table 3.

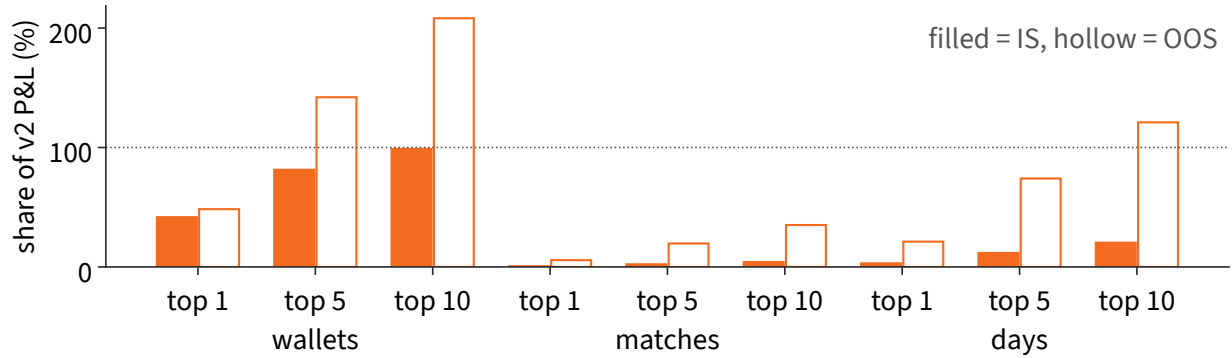


Figure A5. *v2 P&L is concentrated in a few copied wallets.* Share of v2 P&L carried by the top 1, 5 and 10 wallets, matches and days (above 100% means the rest lost money). Source: results/alpha/alpha.json.

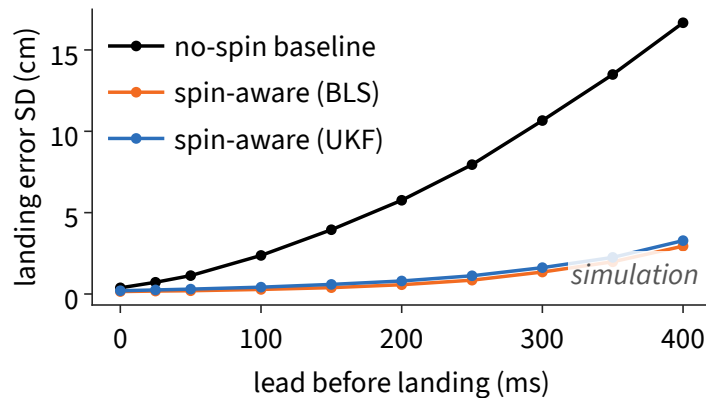


Figure A6. *Simulation: a spin-aware tennis filter cuts landing error.* Landing-point error against lead before the bounce for a 340 fps, 3.6 mm-noise ball tracker (simulation; no tennis video was used). At 200 ms: 5.76 cm without spin, 0.57 cm with it. Source: results/spin/tennis/key_numbers.json.

Table A3. Factor regressions: v2's daily return has no meaningful equity-factor exposure. Fama–French three factors plus momentum, Newey–West 5 lags.

Specification	α %/day (t)	Mkt (t)	SMB (t)	HML (t)	Mom (t)	R^2	days
IS only (headline)	0.80 (8.5)	0.17 (1.4)	0.08 (0.5)	0.04 (0.4)	0.03 (0.4)	3.3%	138
IS, calendar days	0.68 (9.5)	–	–	–	–	2.3%	206
File (5 OOS days)	0.79 (8.5)	0.17 (1.3)	0.09 (0.5)	0.06 (0.6)	0.03 (0.5)	3.3%	142

Notes: the committed file includes five burned-OOS weekdays and is shown only for completeness. Source: results/alpha/alpha.json::C_factor_neutral; French (2026).

Table A4. Variant tally: 4,219 configurations tried; the sensitivities chose nothing.

Family	Count	What it chose
H1-H6 pre-registered and H5/H6 rules	44	frozen primaries (DEVIATIONS D6)
v2 lenses (exit, sizing, selection, cross-market, latency, Kalshi)	3,342	v2 (G_50pct_net100), IS only
v2-safe grid	24	v2-safe (secondary)
tier-0 pre-registered scenarios	432	none (sensitivity grid)
tier-0 v3 in-sample grid	360	frozen v3 rule (blind: FAIL)
maker v1 evaluated variants	8	maker v1 primary (blind: FAIL)
cost-stress cases	4	none (stress)
table-tennis tests TT1-TT5	5	none (untestable)
Total	4,219	
Sensitivity, chose nothing: latency-sweep cells	944	none
Sensitivity, chose nothing: replay cells	36	none

Source: results/paper/variants.json (each count's source file is listed there).

Table A5. Fixed-cost assumptions behind Fig. A1b (\$ per month unless stated).

Item	Low	Central	High	Label
Official live ATP/WTa point-data feed licence	\$1,250	\$5,000	\$10,000	ASSUMPTION
London gateway VPS (AWS eu-west-2, on-demand Linux)	\$34	\$77	\$98	ESTIMATE
Polymarket CLOB / data API	\$0	\$0	\$0	ESTIMATE
HiPerGator (training, backtests)	\$0	\$0	\$0	ESTIMATE
Courtside camera operator (tier-0 only)	\$36	\$36	\$36	ESTIMATE
Cloud GPU for CV inference, per covered event hour (tier-0 only)	\$1	\$1	\$1	ESTIMATE
Operator + GPU hours per covered match (tier-0 only)	\$3	\$4	\$5	ASSUMPTION
Camera position / venue access fee per event (tier-0 only)	\$0	\$250	\$500	ASSUMPTION
120 fps camera kit, amortised over 24 months (tier-0 only)	\$648	\$1,000	\$2,000	ESTIMATE

Notes: no public price exists for an official low-latency tennis feed; the range is anchored to reported figures (bases and URLs in the source file). Source: results/financials/financials.json::cost_assumptions.

Table A6. The latency ladder: method and source of every row of Fig. 1a.

Source	Delay	Method and citation
Our CV engine, call-ready (p50–p99)	4.6–12.2 ms	measured, one L4 GPU, 120 fps held-out video
Our WebRTC → CV pipeline	46 ms p50, 67 ms p99	measured: our own clip over loopback, frame send to call (55 calls; engine fed below real time so it keeps up); not a match feed
Network, Florida → venue gateway	67 ms	measured (live feed p50 65 ms)
In-venue camera	≤0.15 s	physical bound; not available to us
Venue taker order delay	1 s	venue rule (Polymarket, 2026)
Official umpire stamp after the bounce	1–3 s	unmeasured model grid (Regen Sports, 2026; Stats Perform, nd)
Book reprice vs official stamp	–1.2 s	measured median, $n = 482$
Licensed betting video	0.5–8 s	vendor claims, unverified, none for tennis (Stats Perform, nd; NEXT.io, nd; Sportradar, nd; Ably, nd)
Fastest public stream of a listed match	12.2 s	measured, preliminary (no YouTube-derived figure used)
ESPN / PM sports / WTA vs book	28.2 / 30.0 / 44.1 s	measured medians, 2026-10-03 (ESPN, 2026; WTA, 2026)
TV and public streams, general	0.9–60 s	published comparisons (ISPreview, 2021; GL Systemhaus, 2018; The Desk, 2026)

Source: research/v2/latency/results.json, results/tier0/latency_sweep.json::sources, results/home_stream/sub_second_routes.json, results/engine/online_vs_offline.json.

Table A7. Every logged read of held-out data (69 lines of results/oos_peek.log at build time), with its class.

UTC	Class	What was read (truncated)
10-03 10:42	other	OOS prints built
10-03 13:24	non-blind	v2 evaluated on burned OOS (non-blind, labelled)
10-03 13:53	non-blind	v2-causal (D9 fix) evaluated on burned OOS (non-blind)
10-03 17:23	non-blind	tier0 counterfactual: OOS jump table built (burned OOS, non-blind)
10-03 17:29	non-blind	tier0 counterfactual grid evaluated on burned OOS (non-blind, labelled)
10-03 17:31	blind first run	U2 out-of-universe test evaluated (PREREG research/v2/expand, first run)
10-03 17:37	non-blind	v2-safe (lowloss PREREG) evaluated on burned OOS (non-blind, labelled)
10-03 17:37	blind first run	v2-safe U2 out-of-universe test evaluated (PREREG research/v2/lowloss, first run)
10-03 17:43	audit	tier0 independent rebuild: OOS prints read for jump onsets + stale price (burned OOS, non-blind, labelled)
10-03 17:45	audit	tier0 independent rebuild: OOS prints read for jump onsets + stale price (burned OOS, non-blind, labelled)
10-03 17:48	audit	tier0 independent rebuild: OOS prints read for jump onsets + stale price (burned OOS, non-blind, labelled)
10-03 18:00	non-blind	v2 cost stress (fees/costs doubled, book fixed) on burned OOS (non-blind, no parameter change)
10-03 18:06	non-blind	tier0 corrections: OOS post-jump prices built (burned OOS, non-blind, labelled)
10-03 18:13	non-blind	tier0 counterfactual (verifier corrections) evaluated on burned OOS (non-blind, labelled)
10-03 18:27	descriptive	maker venue rules: descriptive in-play book spread/depth + tick changes read from data/live_v2 forward recordings (no strategy, no P&L, no parameter c...
10-03 18:42	non-blind	financials: v2 / v2-safe size-scaled variants (0.5x, 2x, 5x, all prints) evaluated on burned OOS (non-blind, capacity curve only)
10-03 18:47	descriptive	maker live engine plumbing: data/live_v2 clob_17/clob_13 read for message ordering only (price_change vs last_trade_price timestamps, trade dedupe); n...
10-03 18:49	blind first run	maker v1 blind OOS evaluated (first run) (PREREG research/v2/maker/PREREG.md (A); first use of OOS side-market tapes; burned OOS moneyline tapes used ...
10-03 18:49	audit	maker v1 OOS reproduction (recompute committed results.json; no parameter change)
10-03 19:00	live	maker v1 live paper engine REPLAY (recorded live data) on data/live_v2/clob_20261003_1123_20261003_11.jsonl.gz,data/live_v2/clob_20261003_1123_2026100...
10-03 19:13	live	maker v1 live paper TEST run (not the pre-registered session) START: run=20261003T190317Z commit=56a9c8f77b1aaf7b1306c2f1b06d6a2e05c798a8+dirty until=...

UTC	Class	What was read (truncated)
10-03 19:15	audit	P17 clean-clone reproduce.sh (HEAD 56a9c8f, scratch clone, data/linked, backtest workers=1): re-runs the committed frozen scripts incl. run_all.py --...
10-03 19:19	live	maker v1 live paper engine REPLAY (B2: own raw log) on data/live_maker/raw_20261003T190317Z.jsonl.gz window=-+Nonemin clock=recv: run=20261003T191913Z...
10-03 19:20	non-blind	PM review P07: wallet-clustered bootstrap CI of the frozen v2 causal book on burned OOS (non-blind, labelled; the CI the note already quotes, no rule ...
10-03 19:23	blind first run	table tennis TT1-TT4 evaluated (first run)
10-03 19:25	descriptive	TT5 signal decay (scripts/signal_decay.py, commit a5769c786f935dcc05ff2ab90a1947ad0cde4cb5+dirty(signal_decay.py)): reads table tennis UTT OOS prints ...
10-03 19:28	other	table tennis post-hoc diagnostics after TT1-TT4 (closing-print size, fast-tier distance; not pre-registered, no verdict change)
10-03 19:28	live	maker v1 live paper TEST run (not the pre-registered session) START: run=20261003T191915Z commit=ad164da6da24325afcfdbab8fe24dcf5f0516fed+dirty until=...
10-03 19:58	live	maker v1 live paper session (B) START: run=20261003T193534Z commit=114c722e6b203421d8b4f55a5da2b54a1658b658 until=2026-10-04T11:30:00.000Z (PREREG res...
10-03 20:02	audit	audit: independent TT3 reproduction (research/tt/audit_tt3.py; reads UTT IS+OOS tapes/prints, already evaluated in 2195e86) plus labelled post-hoc dia...
10-03 20:03	audit	audit: table tennis checks (research/tt/audit_checks.py): provenance, fees/delays, voids, counts, liquidity; reproduces TT1 closing bins (IS+OOS, alre...
10-03 20:04	audit	audit: financials/risk audit (scripts/financials_audit.py) re-reads the frozen v2 burned-OOS trades and maker blind-OOS fills to reproduce numbers alr...
10-03 20:07	audit	audit: RE-RUN of research/tt/audit_tt3.py with three added post-hoc counts (closest wallet's share of TT2 others' prints, 0-3 s prints at fav >= 0.95 ...
10-03 20:07	audit	audit: RE-RUN of research/tt/audit_tt3.py (traceback location recorded as repo-relative src/v2.py frames only; same reads and numbers, no rule or verd...
10-03 20:09	descriptive	alpha breakdown: descriptive splits of existing v2 trades incl. burned OOS (no parameter change)
10-03 20:10	live	maker v1 live paper engine REPLAY (recorded live data) on data/live_v2/clob_20261003_1123_20261003_1[12].jsonl.gz window=2026-10-03T12:00:00.000Z+60.0...
10-03 20:10	live	maker v1 live paper engine REPLAY (recorded live data) on data/live_v2/clob_20261003_1123_20261003_1[12].jsonl.gz window=2026-10-03T12:00:00.000Z+60.0...

UTC	Class	What was read (truncated)
10-03 20:11	audit	audit: maker v1 blind OOS (A) independent recompute (own code in scratchpad, no import of scripts/maker_oos.py or research/v2/maker/oos_*.py); reads O...
10-03 20:13	live	maker v1 live paper engine REPLAY (recorded live data) on data/live_v2/clob_20261003_1123_20261003_1[123].jsonl.gz window=2026-10-03T13:00:00.000Z+60....
10-03 20:13	live	maker v1 live paper engine REPLAY (recorded live data) on data/live_v2/clob_20261003_1123_20261003_1[1234].jsonl.gz window=2026-10-03T14:00:00.000Z+12...
10-03 20:14	live	maker v1 live paper engine REPLAY (recorded live data) on data/live_v2/clob_20261003_1123_20261003_11.jsonl.gz window=2026-10-03T11:35:00.000Z+25.0min...
10-03 20:14	live	maker v1 live paper engine REPLAY (recorded live data) on data/live_v2/clob_20261003_1123_20261003_1[56789].jsonl.gz window=2026-10-03T16:00:00.000Z+2...
10-03 20:16	non-blind	tier0 latency sweep on burned OOS (non-blind, labelled; no parameter chosen)
10-03 20:16	audit	audit: TT5 same-block split and v2 trade join (research/decay/audit_sameblock.py): re-reads tennis U1 IS + burned OOS prints (non-blind, already evalu...
10-03 20:18	audit	audit: selection check for maker v1 OOS (A): data-api /trades counts and notional (no prices used, no signal, no P&L) for a random sample of 60 OOS si...
10-03 20:21	audit	audit: maker v1 OOS (A) stress (viii) alternative fee reading (makerBaseFee 1000 bps x min(px,1-px), CTF-exchange form) on the already-evaluated book...
10-03 20:22	audit	audit: provenance spot-check of maker v1 OOS cache: re-download 8 cached OOS side tapes (data-api) and 1 Gamma events batch and compare with the cache...
10-03 20:22	live	maker v1 live paper engine REPLAY (B2: own raw log) on data/live_maker/raw_20261003T191915Z.jsonl.gz window=-+Nonemin clock=recv: run=20261003T202228Z...
10-03 20:22	live	maker v1 live paper engine REPLAY (B2: own raw log) on /private/tmp/claude-501/-Users-ojasvamishra32-Library-Application-Support-Claude-scratch-worksp...
10-03 20:24	non-blind	tier0 latency sweep on burned OOS (non-blind, labelled; no parameter chosen)
10-03 20:25	audit	tier0 v3 blind test (a) burned OOS START (scripts/tier0_v3_blind.py; PREREG research/v2/tier0_v3/PREREG.md section 5; non-blind window seen by v1/v2/t...
10-03 20:26	blind first run	tier0 v3 blind test (b) U2 START (scripts/tier0_v3_blind.py; PREREG research/v2/tier0_v3/PREREG.md section 4; first tier-0 table on U2: jump tables + ...
10-03 20:28	blind first run	tier0 v3 blind tests (a)+(b) DONE: burned OOS FAIL, U2-IS FAIL, U2-OOS FAIL -> results/tier0_v3/blind.json

UTC	Class	What was read (truncated)
10-03 20:29	blind first run	tier0 v3 blind tests (a)+(b) DONE: burned OOS FAIL, U2-IS FAIL, U2-OOS FAIL -> results/tier0_v3/blind.json
10-03 20:37	audit	audit: tier-0 v3 blind tests adversarial audit (independent code in scratchpad; COUNTERFACTUAL): re-reads saved v3 burned-OOS/U2 outputs (results/tier...
10-03 20:40	live	maker v1 live paper engine REPLAY (recorded live data) on data/live_v2/clob_20261003_1123_20261003_1[123].jsonl.gz window=2026-10-03T13:00:00.000Z+60....
10-03 20:45	audit	independent rebuild: tier-0 frozen v3 re-implemented in scratchpad (no scripts/tier0_v3_* import; COUNTERFACTUAL) reads U2 prints/universe/raw tapes, ...
10-03 20:46	live	maker v1 live paper engine REPLAY (recorded live data) on data/live_v2/clob_20261003_1123_20261003_1[123].jsonl.gz window=2026-10-03T13:00:00.000Z+60....
10-03 20:52	live	maker v1 live paper session (B) DEVIATION L11-L14 (research/v2/maker/DEVIATIONS_LIVE.md): session run=20261003T193534Z PID 65274 stopped by SIGTERM at...
10-03 21:05	descriptive	alpha breakdown: descriptive splits of existing v2 trades incl. burned OOS (no parameter change)
10-03 20:17	audit	tier0 latency sweep on burned OOS (non-blind, labelled; no parameter chosen) [HiPerGator job 44606812; line written to the cluster copy of this log, s...
10-03 20:36	audit	tier0 latency sweep on burned OOS (non-blind, labelled; no parameter chosen) [HiPerGator job 44608194, the run behind commit f5ff9ff; line written to ...
10-03 20:53	audit	tier0 latency sweep on burned OOS (non-blind, labelled; no parameter chosen) [audit: independent recompute of sweep cells incl. burned OOS video V = 1...
10-03 20:59	audit	tier0 latency sweep on burned OOS (non-blind, labelled; no parameter chosen) [HiPerGator job 44610737, audit re-run adding stamp-lag cells; line writt...
10-03 21:14	live	match replay: forward live recordings of 2026-10-03 used for a labelled 1 s-feed counterfactual replay (no parameter chosen) [logged before the replay...
10-03 21:56	live	match replay: forward live recordings of 2026-10-03 used for a labelled 1 s-feed counterfactual replay (no parameter chosen) [visuals: scripts/match_r...
10-03 22:18	non-blind	capacity study: burned OOS size/cap/coverage grid (non-blind, no parameter chosen) [scripts/capacity_study.py: local v2 1x check, then the HiPerGator ...
10-03 22:24	live	match replay: forward live recordings of 2026-10-03 used for a labelled 1 s-feed counterfactual replay (no parameter chosen) [audit: scripts/match_rep...

10-03 23:01 live match replay selective variant: exploratory, post hoc (forward live recordings of 2026-10-03) [scripts/match_replay_selective.py: the replay restricte...

Table A8. *Pre-registration and deviation trail (git history of the hypothesis, protocol and deviation files).*

Commit	UTC	Subject
7232986	2026-10-03 09:50	Pre-register hypotheses before any results
0f1cc89	2026-10-03 10:15	Data pipeline, Markov model, physics sim, strategies, tier and fast-tier analysis
9d61d1b	2026-10-03 10:32	Freeze all strategy settings before out-of-sample evaluation
69a675f	2026-10-03 10:48	Out-of-sample evaluation, diagnostics, figure fixes
be64ea4	2026-10-03 12:00	Ball tracking results (table tennis H3, tennis broadcast), interactive visual, live-book paper engine
1f229bc	2026-10-03 13:22	Pre-register v2 rule and forward test; add v2 research lenses and verifications
845666f	2026-10-03 13:55	Fix onset hindsight: causal v2; amend forward-test pre-registration before any forward data
e7f34d8	2026-10-03 13:56	Forward window starts 14:00 UTC, after the amended pre-registration
8ede061	2026-10-03 15:00	Pitch deck (12 slides, reviewed); note framing fixes; D10 corrections
337bf7c	2026-10-03 17:12	Pre-register out-of-universe test of frozen v2 (U2)
342d515	2026-10-03 17:30	Pre-register v2-safe (loss-averse, chosen on IS only)
c29734b	2026-10-03 18:26	Pre-register maker v1 blind OOS test and live paper session
0f01362	2026-10-03 18:31	Pre-register table tennis study (TT1-TT5) before fetching tapes
a5769c7	2026-10-03 19:24	Tier-0 CV-edge counterfactual (assumed licensed feed + courtside camera; measured parameters): pre-registration, origina
ba0c3d4	2026-10-03 20:16	Pre-register frozen tier-0 v3 and its blind tests
0e083ed	2026-10-03 21:14	Match replay protocol: selection rule and per-point procedure, written before any P&L